BUS 104: Practice Set — Simulation

Problems and Solutions

Problem 10-17: Vincent Maruggi's Checking Account

Vincent Maruggi, an MBA student at Northern Massachusetts University, has been having problems balancing his checkbook. His monthly income is derived from a graduate research assistantship; however, he also makes extra money in most months by tutoring undergraduates in their decision modeling course. His chances of various income levels are shown here (assume that this income is received at the beginning of each month):

Monthly Income	Probability
\$750	0.40
\$800	0.20
\$850	0.30
\$900	0.10

Maruggi's expenditures also vary from month to month, and he estimates that they will follow this distribution:

Monthly Expenses	Probability
\$ 700	0.10
\$ 800	0.45
\$ 900	0.30
\$1,000	0.15

Maruggi begins his final year with \$1,000 in his checking account. Simulate the cash flow for 12 months and identify Maruggi's (a) ending balance at the end of the year and (b) probability that he will have a negative balance in any month. Replicate your model N times and discuss Maruggi's financial picture based on the average values for these two measures.

Solution to Problem 10-17

1. Probability Distributions and Random Number Intervals

To simulate monthly income and monthly expenses using `RAND()` in Excel, we construct cumulative probability lookup tables that specify the lower limit of each random number interval (`RN intvl LL`):

Income Distribution (`G4:I8`)			Expense Distribution (`G11:I15`)		
Income	**Probability**	**RN Intvl LL**	**Expense**	**Probability**	**RN Intvl LL**
\$750	0.40	0.00	\$ 700	0.10	0.00
\$800	0.20	0.40	\$ 800	0.45	0.10
\$850	0.30	0.60	\$ 900	0.30	0.55
\$900	0.10	0.90	\$1,000	0.15	0.85

Analytically, the expected monthly cash flows are:

- **Expected Income:** \$750(0.40) + \$800(0.20) + \$850(0.30) + \$900(0.10) = \$805.00 per month

- **Expected Expense:** $\$700(0.10) + \$800(0.45) + \$900(0.30) + \$1{,}000(0.15) = \$850.00$ per month
- **Expected Net Monthly Cash Flow:** $\$805.00 - \$850.00 = -\$45.00$ per month
- **Theoretical Expected Year-End Balance:** $\$1{,}000 + 12(-\$45.00) = \$460.00$

2. Spreadsheet Simulation Model Logic (see `P10-17.xls`)

For each month $t = 1, 2, \ldots, 12$ (rows `4:15`), the checkbook balance evolves according to:

$$\text{Ending Balance}_t = \text{Beginning Balance}_t + \text{Income}_t - \text{Expense}_t$$

where $\text{Beginning Balance}_1 = \$1{,}000$ (cell `I17`) and $\text{Beginning Balance}_t = \text{Ending Balance}_{t-1}$ for $t \geq 2$.

- **Beginning Balance (`B4`, `B5:B15`):** `=I17` in Month 1 (`B4`), and `=E4` copied down in Months 2–12 (`B5:B15`).
- **Simulated Income (`C4:C15`):** `=LOOKUP(RAND(), $I$5:$I$8, $G$5:$G$8)`
- **Simulated Expense (`D4:D15`):** `=LOOKUP(RAND(), $I$12:$I$15, $G$12:$G$15)`
- **Ending Balance (`E4:E15`):** `=B4+C4-D4`
- **(a) Year-End Balance (`E16`):** `=E15` (the ending balance in Month 12)
- **(b) Prob. of Negative Balance (`E17`):** `=COUNTIF(E4:E15,"<0")/COUNT(E4:E15)` (the fraction of the 12 months in which the ending balance is strictly negative)

3. Sample Single-Year Replication and 200-Replication Summary (`P10-17.xls`)

Sample 12-Month Trial (`A3:E17`)				
Month	**Begin Bal**	**Income**	**Expense**	**End Bal**
1	\$1,000	\$850	\$ 900	\$ 950
2	\$ 950	\$900	\$ 800	\$1,050
3	\$1,050	\$750	\$1,000	\$ 800
4	\$ 800	\$750	\$ 800	\$ 750
5	\$ 750	\$750	\$ 900	\$ 600
6	\$ 600	\$800	\$ 800	\$ 600
7	\$ 600	\$800	\$ 800	\$ 600
8	\$ 600	\$800	\$ 900	\$ 500
9	\$ 500	\$750	\$ 800	\$ 450
10	\$ 450	\$750	\$ 900	\$ 300
11	\$ 300	\$850	\$ 800	\$ 350
12	\$ 350	\$750	\$ 800	\$ 300

Summary Statistics	
Performance Measure	**Value**
Single 12-Month Trial (`E16:E17`):	
Year-end balance (`E16`)	\$300.00
Prob. of neg. balance (`E17`)	0.00%
200 Replications (`Data Table K3:M203`):	
Average ending bal (`I20`)	**\$447.75**
Prob(neg balance) (`I21`)	**0.0208 (2.08%)**
Theoretical Benchmark:	
Expected monthly net flow	−\$45.00
Expected year-end balance	\$460.00

4. Discussion of Maruggi's Financial Picture

- **(a) Average Ending Balance:** Across $N = 200$ simulated years (`L4:L203`), Maruggi's average year-end balance is **\$447.75** (`=AVERAGE(L4:L203)`), which is close to the theoretical expected year-end balance of \$460.00.
- **(b) Probability of a Negative Monthly Balance:** Across the 200 replications (`M4:M203`), the average probability of having a negative balance in a given month is **0.0208** (or **2.08%**, representing 50 negative-balance months out of $200 \times 12 = 2{,}400$ total simulated months). In addition, 19 of the 200 simulated years (9.5%) experienced at least one negative monthly balance, and 17 of the 200 runs (8.5%) ended Month 12 in the red.
- **Financial Assessment:** Although Maruggi's average monthly expenses (\$850) exceed his average monthly income (\$805) by \$45 per month, his \$1,000 starting balance provides a sufficient cushion to carry him through his final 12-month academic year with very low risk (2.08% in any given month) of overdrawing his checking account. However, because his net cash flow is negative on average, he cannot sustain this spending pattern indefinitely without depleting his savings once the year ends.

Problem 10-37: Clemson Police Department Solicitations

The Clemson Police Department makes annual door-to-door solicitations for funds. Residents of each visited house are asked to contribute either \$15 (and receive a free family portrait package) or \$25 (and receive two free family portrait packages). An analysis from previous years' solicitations indicates that

- only 80% of the homes visited have the man or woman of the house at home.
- when someone is home, there is only a 40% chance that he or she will make a donation.
- of the people making donations, there is a 50% chance they will contribute \$15 and a 45% chance they will contribute \$25. Occasionally (5% chance), a person makes a donation in excess of \$25. Such distributions follow a discrete uniform distribution between \$30 and \$50 (in increments of \$1).

The police chief plans to visit 30 houses tomorrow. Set up a simulation model and replicate it N times to determine the probability that the chief will receive more than \$250 in donations from these 30 houses.

Solution to Problem 10-37

1. Probability Structure and Lookup Table Setup

At each visited house $i \in \{1, 2, \ldots, 30\}$, three sequential random events determine the donation received:

1. **Resident at Home (`Prob of home` = 0.80 in `K9`):** Bernoulli trial with success probability 0.80.
2. **Willing to Donate (`Prob of donate` = 0.40 in `K10`):** Conditional on someone being home, a Bernoulli trial with success probability 0.40. Thus, the unconditional probability that a visited house makes a donation is $0.80 \times 0.40 = 0.32$ (32%).
3. **Donation Amount Given a Donor (`I15:M38`):**
   - \$15 with probability 0.50 (random number interval [0.00, 0.50))
   - \$25 with probability 0.45 (random number interval [0.50, 0.95))
   - $\$30, \$31, \ldots, \$50$ (21 integer values) sharing the remaining 0.05 probability equally, so each integer amount from \$30 to \$50 has probability $\frac{0.05}{21} \approx 0.002381$ (in `J18:J38`, entered as `=0.05/21`).

Analytically, the expected donation given that a resident donates is:

$$E[\text{Donation} \mid \text{Donor}] = \$15(0.50) + \$25(0.45) + \$40(0.05) = \$7.50 + \$11.25 + \$2.00 = \$20.75$$

Across 30 houses, the expected number of donors is $30 \times 0.32 = 9.6$ donors, and the theoretical expected total donation is $30 \times 0.32 \times \$20.75 = \$199.20$.

2. Spreadsheet Simulation Model Logic (see `P10-37_Excel.xls`)

Donor Amount Lookup Table (`I15:M38`)			
Donation $ (`I`)	**Probability (`J`)**	**Lineup / RN LL (`L`)**	**Donation $ (`M`)**
\$15	0.5000	0.0000	\$15
\$25	0.4500	0.5000	\$25
\$30	$0.05/21 \approx 0.0024$	0.9500	\$30
\$31	$0.05/21 \approx 0.0024$	0.9524	\$31
⋮	⋮	⋮	⋮
\$50	$0.05/21 \approx 0.0024$	0.9976	\$50

For each of the 30 houses in rows `4:33` (`A4:A33` $= 1, \ldots, 30$):

- **Home? (`B4:B33`):** `=IF(RAND()<$K$9, 1, 0)` (equals 1 if someone is home, 0 otherwise)
- **Donate? (`C4:C33`):** `=IF(RAND()<$K$10, 1, 0)` (equals 1 if willing to donate, 0 otherwise)
- **Donation Amount (`D4:D33`):** `=VLOOKUP(RAND(), $L$16:$M$38, 2)` (Alternatively, one could use `RANDBETWEEN(30,50)` when the third branch is triggered.)
- **Actual Outcome (`E4:E33`):** `=B4*C4*D4` (actual dollars collected from house i)
- **Total Donation (`E35`):** `=SUM(E4:E33)`
- **Donation** $>$ \$250**? (`E36`):** `=IF(E35>250, 1, 0)` (indicator equal to 1 if total donations exceed \$250)

3. Simulation Results Across 200 Replications (`P10-37_Excel.xls`)

A two-output Data Table in `O3:Q203` replicates the 30-house solicitation $N = 200$ times, recording the total donation (`P4:P203`, linked to `=E35` in `P3`) and the indicator for exceeding \$250 (`Q4:Q203`, linked to `=E36` in `Q3`):

Sample Trial (First 10 of 30 Houses)					Simulation Summary	
House	**Home?**	**Donate?**	**Amount**	**Outcome**	**Measure**	**Value**
1	1	0	\$25	\$ 0	*Saved Single Trial (`E35:E36`):*	
2	1	0	\$46	\$ 0	Total donation (`E35`)	\$150.00
3	1	0	\$25	\$ 0	Exceeds \$250? (`E36`)	0 (No)
4	1	0	\$15	\$ 0		
5	0	1	\$15	\$ 0	*200 Replications (`O3:Q203`):*	
6	1	0	\$25	\$ 0	Average total donation	\$203.14
7	1	0	\$25	\$ 0	Runs with donation $>$ \$250	38 of 200
8	1	0	\$15	\$ 0	**Prob($>$ \$250) (`U4`)**	**0.19 (19.0%)**
9	1	1	\$25	\$25		
10	1	0	\$15	\$ 0	*Theoretical Benchmark:*	
⋮	⋮	⋮	⋮	⋮	Expected total donation	\$199.20

- **Conclusion:** Based on 200 replications (`U4` = `=AVERAGE(Q4:Q203)`), the estimated probability that the police chief will receive more than \$250 in donations from visiting 30 houses is **0.19** (or **19.0%**). The simulated average collection across the 200 trials is \$203.14, closely matching the theoretical expectation of \$199.20.

Problem 10-46: Management Resources, Inc. (Project Scheduling)

Tom Schriber, director of personnel of Management Resources, Inc., is in the process of designing a program that its customers can use in the job-finding process. Some of the activities include preparing resumés, writing letters, making appointments to see prospective employers, researching companies and industries, and so on. Information on the activities is shown in the following table:

	Days		
Activity	**Mean**	**Standard Deviation**	**Immediate Predecessors**
A	10.00	0.67	—
B	7.17	0.50	—
C	3.17	0.17	—
D	20.00	3.33	A
E	7.00	0.33	C
F	10.00	0.33	B, D, E
G	7.33	0.67	B, D, E
H	15.00	0.33	F
I	11.17	0.50	F
J	7.00	0.33	G, H
K	6.67	0.67	I, J
L	2.17	0.50	G, H

Assume that all activity durations follow a normal distribution, with the means and standard deviations shown. Round off all activity times to two decimal points.

Use simulation to determine the average project completion time and the probability that the project will take at least 75 days.

Solution to Problem 10-46

1. Project Network Logic and Spreadsheet Model Setup (see `P10-46.xls`)

Each activity $i \in \{\text{A}, \text{B}, \ldots, \text{L}\}$ has a normally distributed duration $T_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ rounded to two decimal places:

- **Simulated Activity Time (`E4:E15`):** `=ROUND(NORMINV(RAND(), C4, D4), 2)` (or `=ROUND(NORM.INV(RAND(), C4, D4), 2)` in modern Excel).
- **Start Time (`F4:F16`):** An activity cannot start until all of its immediate predecessors have finished. Thus, the start time of an activity with no predecessors is 0, and the start time of any other activity is the maximum of the finish times of its immediate predecessors.
- **Finish Time (`G4:G16`):** `=F4+E4` (Start Time + Activity Time).
- **Project Completion Time (`F16`, `G16`, and `J5`):** A dummy `End` row (`Row 16`) with immediate predecessors K and L sets `F16` = `=MAX(G14,G15)` and `G16` = `=F16+E16` (with `E16` = 0), representing the overall project finish time.

Note: In `P10-46.xls`, the means and standard deviations in `C4:D15` are stored as exact underlying fractions (e.g., $7\frac{1}{6} = 7.1667$, $\frac{2}{3} = 0.6667$, $3\frac{1}{3} = 3.3333$) and displayed to two decimal places.

2. Activity Schedule Table and Formulas (`A3:G16` in `P10-46.xls`)

Act.	Immed. Pred.	Mean	Std. Dev.	Act. Time	Start	Finish	Start Time Formula (`Col F`)
A	—	10.00	0.67	10.15	0.00	10.15	`0`
B	—	7.17	0.50	6.35	0.00	6.35	`0`
C	—	3.17	0.17	2.69	0.00	2.69	`0`
D	A	20.00	3.33	19.83	10.15	29.98	`=G4`
E	C	7.00	0.33	7.14	2.69	9.83	`=G6`
F	B, D, E	10.00	0.33	9.97	29.98	39.95	`=MAX(G5,G7,G8)`
G	B, D, E	7.33	0.67	6.79	29.98	36.77	`=MAX(G5,G7,G8)`
H	F	15.00	0.33	15.10	39.95	55.05	`=G9`
I	F	11.17	0.50	11.54	39.95	51.49	`=G9`
J	G, H	7.00	0.33	6.94	55.05	61.99	`=MAX(G10,G11)`
K	I, J	6.67	0.67	6.20	61.99	68.19	`=MAX(G12,G13)`
L	G, H	2.17	0.50	2.17	55.05	57.22	`=MAX(G10,G11)`
End	K, L	0.00	0.00	0.00	**68.19**	**68.19**	`=MAX(G14,G15)`

3. Simulation Results Across 200 Replications (`P10-46.xls`)

Using a Data Table in `L2:M202` to replicate the project network $N = 200$ times:

- **Single-Replication Completion Time (`J5`):** `=G16` equals **68.19 days**.
- **Average Project Completion Time (`J8`):** `=AVERAGE(M3:M202)` equals **68.52 days** (68.5188 days, with simulated completion times ranging from 59.76 days to 82.49 days across the 200 runs).
- **Probability Project Takes at Least 75 Days (`J9`):**
  - In `P10-46.xls`, cell `J9` uses `=1-COUNTIF(M3:M202,"<=75")/200`, which yields **0.035** (or **3.5%**, corresponding to 7 out of 200 runs strictly exceeding 75 days).
  - Because one replication (Run 196 in `M198`) finished in exactly 75.00 days, counting runs with completion time ≥ 75 days via `=COUNTIF(M3:M202,">=75")/200` yields $8/200 =$ **0.040** (or **4.0%**).
- **Comparison with Analytical PERT Approximation:** Along the dominant path A $\to$ D $\to$ F $\to$ H $\to$ J $\to$ K, the sum of mean durations is $10.00 + 20.00 + 10.00 + 15.00 + 7.00 + 6.67 = 68.67$ days, and the path standard deviation is

$$\sigma_{\text{path}} = \sqrt{(2/3)^2 + (10/3)^2 + (1/3)^2 + (1/3)^2 + (1/3)^2 + (2/3)^2} = \sqrt{111/9} \approx 3.512 \text{ days.}$$

  Under the normal approximation, $P(T \geq 75) = 1 - \Phi\left(\frac{75-68.67}{3.512}\right) = 1 - \Phi(1.80) \approx 0.0359$ (3.59%), which agrees closely with the simulated probability of 3.5%–4.0%.